Summary in Graph

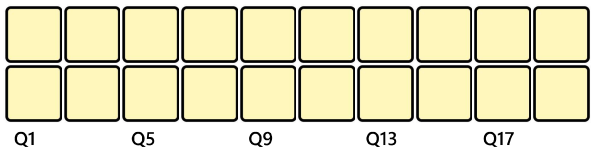
Exam Summary (GO Classes CS/DA Test Series 2026 | Probability | Topic Wise Test)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	20 (5 + 15)
Total Marks:	35 (5 + 30)
Exam Duration:	60 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (107) out of 1069 users

EXAM RESPONSE | EXAM STATS | FEEDBACK

Technical

Q #1 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

Two fair six-sided dice are rolled simultaneously. Let the random variable X denote the sum of the numbers appearing on the top faces of the two dice.

Which of the following correctly represents the probability distribution of X ?
Let $x_k = k$ denote the possible values of X, and $p_k = P(X = x_k)$ denote the probability that the sum equals x_k.

A. $x_k = k, p_k = 1/6, k = 1, 2, \cdots, 6.$
B. $x_k = k, p_k = 1/11, k = 2, 3, \cdots, 12.$
C. $x_k = k, p_k = \min[k-1, 13-k]/36, k = 2, 3, \cdots, 12.$
D. $x_k = k, p_k = k/48, k = 2, 3, \cdots, 12.$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

Among 35 students in a class, 17 earned "A" on the midterm, 14 earned "A" on final exam, and 11 did not earn "A" on either exam. What is the probability that a randomly selected student from this class earned "A" on both exams?

A. $7/35$
B. $24/35$
C. $17/35$
D. $14/35$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Numerical Type | Award: 1 | Penalty: 0 | Probability

Let X be a random variable with probability density:

$$f(x) = \begin{cases} c\left(1 - x^2\right), & -1 < x < 1 \\ 0, & \text{otherwise} \end{cases}$$

The value of c is _____

Your Answer: | Correct Answer: 0.75 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

Suppose that we toss a coin having a probability p of coming up heads, until the first head appears. Letting N denote the number of flips required, then assuming that the outcome of successive flips are independent, N is a random variable taking on one of the values $1, 2, 3, \ldots,$
Which of the following is correct?

A. $P\{N = n\} = (1-p)^{n-1}p; n \geq 1$
B. $P\{N = n\} = (1-p)^{n}p; n \geq 1$
C. $P\{N = n\} = p^{n-1}(1-p); n \geq 1$
D. $P\{N = n\} = p^{n}(1-p); n \geq 1$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

Suppose, two events A and B are independent then which of the following is/are correct?
$P(.)$ stands for probability

1. $P(A \cap B) = P(A)P(B)$
2. $P(A \cup B) = P(A) + P(B) - P(A)P(B)$
3. $P(A \mid B) = 0$
4. $P(A \mid \neg B) = P(A)$

A. $1, 2$
B. $1, 2, 3$
C. $1, 2, 4$
D. $1, 2, 3, 4$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Suppose $X = 0$ with probability $\frac{1}{2}$, 1 with probability $\frac{1}{4}$, 2 with probability $\frac{1}{8}$, and more generally n with probability $1/2^{n+1}$. This is an example where X can take infinitely many values (although still countably many values). What is the expectation of X?

A. $1/2$
B. 1
C. 2
D. $3/2$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Numerical Type | Award: 2 | Penalty: 0 | Probability

Three candidates - A, B, and C - are contesting an election. Exactly one of them will win.
Let:

- $\Omega = \{A, B, C\}$ be the sample space,
- $P(A), P(B)$, and $P(C)$ denote the probabilities that A, B, and C win, respectively.

You are given:

- $P(A) = P(B)$,
- $P(C) = \frac{1}{2} \cdot P(A)$
  (i.e., C is half as likely to win as A or B)

Let $E = \{A, C\}$.

Find: $P(E)$

Your Answer: | Correct Answer: 0.6 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Consider a coin with probability p to be heads. Determine the probability that the first heads will appear on the even-numbered tosses.

A. $\frac{2-p}{1-p}$

B. $\frac{1-p}{2-p}$

C. $\frac{p}{2-p}$

D. $\frac{1}{2-p}$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Suppose, two events A and B are mutually exclusive then which of the following is/are correct? $P(.)$ stands for probability.

1. $P(A \cap B) = 0$
2. $P(A \cup B) = P(A) + P(B)$
3. $P(A \mid B) = 0$
4. $P(A \mid \neg B) = \frac{P(A)}{1-P(B)}$

A. $1, 2$
B. $1, 2, 3$
C. $1, 3, 4$
D. $1, 2, 3, 4$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Select Type | Award: 2 | Penalty: 0 | Probability

Roll a die.

- Let A be the event that the outcome on the die is an even number.
- Let B be the event that the outcome on the die is 4 or smaller.
- Let C be the event that the outcome on the die is 3 or larger.

Choose the correct option(s)?

A. A and B are dependent events
B. A and B are independent events
C. B and C are dependent events
D. B and C are independent events

Your Answer: | Correct Answer: B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

It is estimated that 50% of emails are spam emails. Some software has been applied to filter these spam emails before they reach your inbox. A certain brand of software claims that it can detect 99% of spam emails, and the probability for a false positive (a non-spam email detected as spam) is 5%.

Now if an email is detected as spam, then what is the probability that it is in fact a non-spam email?

A. $1/104$
B. $95/104$
C. $5/104$
D. $99/104$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Select Type | Award: 2 | Penalty: 0 | Probability

Which of the following is/are true?

A. If $P(A \mid B) = 0.5$ and $P(B) = 0.5$, then the events A and B are necessarily independent.
B. If two events A and B are independent then both $P(A \mid B) = P(A)$ and $P(B \mid A) = P(A)$.
C. If $P(A \cap B) \geq 0.10$ then $P(A) \geq 0.10$.
D. If $P(A) > P(B)$, and $P(C \mid A) > P(C \mid B)$, then $P(A \mid C) > P(B \mid C)$.

Your Answer: | Correct Answer: C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Multiple Select Type | Award: 2 | Penalty: 0 | Probability

Which of the following is/are true?

A. The formula for the variance $\mathrm{Var}(X+Y) = \mathrm{Var}(X) + \mathrm{Var}(Y)$ works regardless of whether the RV's X and Y are independent or not.
B. $E(X-Y) = E(X) - E(Y)$ for any R.V.s X and Y, regardless of whether they are independent or not.
C. $\mathrm{Var}(X-Y) = \mathrm{Var}(X) - \mathrm{Var}(Y)$ for any independent R.V.s X and Y.
D. For any R.V X, we have $\mathrm{Var}(aX) = a^2\,\mathrm{Var}(X)$.

Your Answer: | Correct Answer: B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Multiple Select Type | Award: 2 | Penalty: 0 | Probability

Which of the following is/are true?

A. It is always true that the standard deviation is less than the variance
B. For any two independent random variables X, Y on the same probability space, if $\mathrm{E}(X) > \mathrm{E}(Y)$ then $\Pr[X > Y] > 0$
C. Let A and B be events in the same sample space. If $p(A \mid B) = 1/2$, then $p\left(A \mid B^c\right) = 1/2$.
D. For every random variable X, $E\left(X^2\right) \geq E(X)^2$.

Your Answer: | Correct Answer: B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Suppose that X is a random variable where: $P(X=1) = 1/2 \quad P(X=2) = 1/4 \quad P(X=4) = 1/4$.
What is the value of $P\left(X^2 - 3X + 2 = 0\right)$?

A. $3/4$
B. $1/4$

C. $1/2$
D. $1/3$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #16 | Multiple Select Type | Award: 2 | Penalty: 0 | Probability

For each of the following relations, determine which is TRUE for arbitrary events A, B, and C. (Note: to be true "for arbitrary events", it must be true for any such event. Use a Venn diagram if it is helpful.)

A. $(A \cup B \cup C)^c = A^c \cup B^c \cup C^c$
B. $(A \cup B) \cap (A^c \cup B^c) = (A \cap B^c) \cup (A^c \cap B) \cup (A^c \cap B \cap C^c)$
C. $(A \cap B) \cup (A \cap B^c) \cup (A^c \cap B) = (A^c \cap B^c)^c$
D. $(A - B) - C = A - (B - C)$

Your Answer: | Correct Answer: B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #17 | Multiple Select Type | Award: 2 | Penalty: 0 | Probability

Which of the following is/are true?

A. If two events A and B are disjoint, they must also be independent.
B. If two events A and B are collectively exhaustive then $P(A) + P(B) = 1$.
C. If two events A and B are collectively exhaustive then $P(A \cup B) = 1$.
D. If two evens A and B are independent, then A^c and B^c are also independent.

Your Answer: | Correct Answer: C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #18 | Multiple Select Type | Award: 2 | Penalty: 0 | Probability

Which of the following is/are true?

A. Disjoint events are independent.
B. If events A and B are independent, then conditioned on any other event C, A and B are still independent.
C. $P(A^c \cup B) \leq 1 - P(A) + P(B)$ for any A and any B.
D. If A and B are independent, and B and C are independent, then A and C are also independent.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #19 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

A coin is tossed until a head appears two times in a row. Given that we are using a fair coin, what is the probability that we toss the coin exactly 4 times such that the two consecutive heads are the 3rd and 4th trials?

A. $\frac{1}{4}$
B. $\frac{1}{8}$
C. $\frac{1}{2}$
D. $\frac{1}{6}$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #20 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Suppose X is a random variable taking values in $\{-2, -1, 0, 1, 2, 3, 4, 5\}$, each with probability $\frac{1}{8}$. Let $Y = X^2$. Find $E[Y]$.

A. $12/8$
B. $60/8$
C. 0
D. $1/8$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 05sec | Discuss

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